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ANISHA BANDYOPADHYAY

Physical Chemist | Raman Spectroscopy | Non-linear spectroscopy | Molecular Dynamic Simulations

EDUCATION

Ph.D. in Chemistry (*Expected 2026*)

Bhabha Atomic Research Centre /
Homi Bhabha National Institute

Research Area

Raman Spectroscopy, Nonlinear-Spectroscopy, Hydration-shell Chemistry, Complex interfacial chemistry, Molecular level interaction

M.Tech. | Materials Science & Engineering

Indian Institute of Technology
Kharagpur (CGPA: 9.56/10)

CORE COMPETENCIES

- Linear and non-linear Spectroscopy
- Laser alignment
- Experimental Design
- Data Interpretation
- Scientific Problem Solving
- Instrument Development

Professional Profiles

LinkedIn: [linkedin.com/in/anishachem633/](https://www.linkedin.com/in/anishachem633/)

Github: [anisha633.github.io](https://github.com/anisha633)

Google Scholar: [Anisha Bandyopadhyay - Google Scholar](#)

Reference

Available upon request.

PROFESSIONAL SUMMARY

Experimental Physical Chemist with 5 years of experience developing and applying advanced optical spectroscopy techniques to investigate molecular interactions and interfacial phenomena. Hands-on expertise in designing laser-based Raman spectroscopy systems, nonlinear optical spectroscopy (VSFG), molecular dynamics simulations, and advanced spectral analysis. Proven record of developing custom experimental platforms, publishing 11 peer-reviewed research articles, and presenting research internationally. Seeking to apply expertise in spectroscopy, analytical instrumentation, and scientific problem-solving to industrial R&D.

RESEARCH HIGHLIGHTS (RECENT)

- Mechanistic interpretation of the complexation reactions of Selenium oxyanions, toxic ground water contaminant, at the air/water-Goethite (Mineral of iron hydroxyl oxide) interfaces using state-of art interface selective VSFG spectroscopy.
- Investigated the competing roles of hydrophobic and electrostatic interactions governing bile salt adsorption at phospholipid membranes using interface-selective VSFG spectroscopy, providing molecular insights relevant to drug delivery systems.
- Developed a custom polarization- and temperature-dependent Raman spectroscopy platform for investigating weak hydrogen-bonding interactions through polarization-resolved measurements.

COMPLETED Projects

- Preparation of doped Ceria samples via gel combustion method for Oxygen evolution reaction (OER) using XRD, Raman and PALS
- Performed quantitative analysis and interpretation of structural, spectroscopic, and gas sensing data for CuO particulate thick films to evaluate ethanol sensing performance.

INDUSTRY RELEVANT EXPERIENCES

- Aligned, calibrated, and maintained laser-based optical systems.
- Maintenance and troubleshooting of Raman instrumentation.
- Developed and validated spectroscopic methods for quantitative molecular analysis.
- Authored technical reports, research publications, and experimental protocols

TECHNICAL EXPERTISE

Spectroscopy

- Vibrational Sum Frequency Generation
- Polarized and Temperature dependent Raman
- Positron Annihilation Lifetime Spectroscopy (PALS)
- FTIR

Instrumentation

- Laser and optical alignment
- Home built Raman set-up

Characterization

- XRD
- FTIR
- SEM
- Raman
- Tensiometer
- DLS

Software

- GROMACS
- TURBOMOLE
- OriginPro
- Igor
- MS Office
- Linux